

Bachelor's Degree in Software Engineering

Data Visualization

2025-2026

Python Project

- **Practice conditions:**

- The assignment is **individual**. If similar code fragments are detected (similar errors, similar problem-solving approach, or similarities with content available on the internet), the assignment will receive a **grade of zero**.
- The deliverable for this assignment consists of the following elements:
 - **Python code** in an **.ipynb file (Jupyter Notebook)**, preferably executable. This file should include the code and the necessary Markdown to provide context for the project.
 - **A PDF document (report)** containing:
 - Student name
 - Questions about the dataset to be answered using graphs
 - Screenshots of the charts
 - AI usage declaration (if applicable)
 - Bibliography /conclusions
 - Summary tables (see example below) explaining how the requirements have been met or confirming their fulfillment.
 - **A link to a video of maximum 5 minutes**, hosted in a repository (OneDrive, YouTube, Dropbox, etc.).
Do NOT upload the video to Blackboard.
 - **Optional files:** CSV, JSON, or Parquet files.
Only **one CSV file may be stored locally**. If more are needed, they must be uploaded to a repository or website such as GitHub.
- **Deliverable date will be agreed.**

VERY IMPORTANT: DO NOT upload a ZIP file. Each file must be uploaded separately.

- The executable elements of the assignment must include **all necessary commands** to run on any Python installation.
If additional libraries beyond the standard ones are used, the files must include the installation and loading of these packages.
If the libraries are already installed, the code **should not attempt to reinstall them**.

▪

- If the code **does not run or fails to compile**, the grade will be **0**.
- Students are encouraged to complete the assignment using **cheat sheets and their notes**, as preparation for the exam.

- **Assignment statement:**

- You are required to create a **Jupyter Notebook (JN)** (It can be also a **.py**) that meets the following requirements:
 - The Jupyter Notebook must include **Markdown cells and code comments** that help explain the **storytelling behind the visualizations**.
 - **Requirements:**
 - Choose a **dataset that has not been selected by another student**. To avoid duplication, students must **post in the forum which dataset they plan to use**.
 - It may be a dataset used in the **final degree project (TFG)**.
 - Keep in mind that **another dataset will be required in the next unit**.
 - The dataset can be **validated with the professor if necessary**.
 - Include **at least 3 questions**, which will be answered using visualizations.
 - Include **at least 5 visualizations**, using **at least 3 different types**.
 - These visualizations must follow the **best practices studied in Unit 1**.
 - Use **at least 3 different libraries**, chosen from: Matplotlib, ipywidgets, Plotly, Bokeh, Jupyter Dash, Seaborn, etc.
 - Use **at least 2 different data sources**, such as: CSV, API, pandas, parquet, etc.
 - The submission must be **fully reproducible when executed**. Therefore, the Jupyter Notebook must include:
 - library imports
 - file access instructions
 - API access explanations (if applicable)
 - required files if necessary
 - Submitting examples **taken directly from a website** will be penalized with a **grade of 0**, as it will be considered **plagiarism**.
 - Finally, you must create a **video of no more than 5 minutes** explaining the **questions and answers using graphs**, using a **storytelling style**. This is **NOT** a recorded **Python class or explanation of its libraries**.

Examples of summary tables:

Question	Charts	Chart type	Library
Question 1 XXXXX	Image 1	Violin Plot	Matplotlib Ipywidgets
	Image 2	histogram	
Question 2 XXXXX	Image 3	Line chart	Plotly
Question 3	Image 4	subplot 1 bar chart	Seaborn
	Image 5	Subplot 2 bar chart	

List of deliverables	Done/ Not None
Code	
PDF (student info, questions, screenshots, declaration of use of AI, bibliography & conclusions, and summary table)	
Link video	
Optional files	